

Web Programming Lab

Week 5

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Batch A2

Lab Exercises:

Common [urls.py](#) (Included in project urls):

```
from django.urls import path
from . import views

urlpatterns = [
    path('lab5/q1/', views.student_view, name='student_view'),
    path('lab5/q2/', views.employee_view, name='employee_view'),
]
```

Q1. Create a web form which allows the teacher to enter the student details like name, date of birth, address, contact number, email id and marks of English, Physics and Chemistry. After filling all the information and clicking the submit button, details should be added to a textarea displayed on the same page. Display the total percentage of marks obtained in a label.

Template:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Student Details Entry</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap
.min.css" rel="stylesheet">
    <style>
        body { background-color: #f8f9fa; }
        .card { box-shadow: 0 4px 6px rgba(0,0,0,0.1); }
        .result-area { background-color: #e9ecef; }
    </style>
</head>
<body>

<nav class="navbar navbar-expand-lg navbar-dark bg-dark mb-4">
    <div class="container">
        <a class="navbar-brand" href="#">Lab 5</a>
        <div class="navbar-nav">
            <a class="nav-link" href="/lab5/q1/">Q1: Student Form</a>
            <a class="nav-link" href="/lab5/q2/">Q2: Promotion Check</a>
        </div>
    </div>
</nav>

<div class="container py-5">
    <div class="row justify-content-center">
        <div class="col-lg-8">
            <h2 class="text-center mb-4 text-primary">Student Records
System</h2>

            <div class="card">
```

```

<div class="card-header bg-primary text-white">
    <h5 class="mb-0">Enter Student Details</h5>
</div>
<div class="card-body">
    <form method="POST">
        {% csrf_token %}

        <div class="row mb-3">
            <div class="col-md-6">
                <label for="name"
class="form-label">Student Name</label>
                <input type="text"
class="form-control" name="name" required>
            </div>
            <div class="col-md-6">
                <label for="dob"
class="form-label">Date of Birth</label>
                <input type="date"
class="form-control" name="dob" required>
            </div>
        </div>

        <div class="mb-3">
            <label for="address"
class="form-label">Address</label>
            <textarea class="form-control"
name="address" rows="2" required></textarea>
        </div>

        <div class="row mb-3">
            <div class="col-md-6">
                <label for="contact"
class="form-label">Contact Number</label>
                <input type="tel"
class="form-control" name="contact" required>
            </div>
            <div class="col-md-6">
                <label for="email"
class="form-label">Email ID</label>
                <input type="email"
class="form-control" name="email" required>
            </div>
        </div>

        <hr class="my-4">

```

```

        <h6 class="text-secondary mb-3">Academic
Marks</h6>

        <div class="row mb-4">
            <div class="col-md-4">
                <label for="english"
class="form-label">English</label>
                <input type="number"
class="form-control" name="english" min="0" max="100" required>
            </div>
            <div class="col-md-4">
                <label for="physics"
class="form-label">Physics</label>
                <input type="number"
class="form-control" name="physics" min="0" max="100" required>
            </div>
            <div class="col-md-4">
                <label for="chemistry"
class="form-label">Chemistry</label>
                <input type="number"
class="form-control" name="chemistry" min="0" max="100" required>
            </div>
        </div>

        <div class="d-grid gap-2">
            <button type="submit" class="btn
btn-primary btn-lg">Submit Details</button>
        </div>
    </form>
</div>

<div class="card mt-4">
    <div class="card-header bg-secondary text-white">
        <h5 class="mb-0">Submission Output</h5>
    </div>
    <div class="card-body">
        <div class="mb-3">
            <label class="form-label fw-bold">Formatted
Details:</label>
            <textarea class="form-control result-area"
rows="5" readonly>{{ details_log }}</textarea>
        </div>
    </div>
</div>

```

```

                                <div class="alert alert-info d-flex
align-items-center" role="alert">
                                <span class="me-2">Total Percentage:</span>
                                <span class="badge bg-success fs-5">
                                    {% if percentage %}
                                        {{ percentage }}%
                                    {% else %}
                                        0%
                                    {% endif %}
                                </span>
                                </div>
                            </div>
                        </div>
                    </div>

                </div>
            </div>
        </div>

    </body>
</html>

```

Backend:

```

def student_view(request):
    context = {}
    if request.method == 'POST':
        name = request.POST.get('name')
        eng = int(request.POST.get('english', 0))
        phy = int(request.POST.get('physics', 0))
        chem = int(request.POST.get('chemistry', 0))

        total = eng + phy + chem
        percentage = round((total / 300) * 100, 2)

        details_str = (
            f"Name: {name}\n"
            f"Address: {request.POST.get('address')}\n"
            f"Email: {request.POST.get('email')}\n"
            f"Marks: Eng({eng}), Phy({phy}), Chem({chem})"
        )

        context['details_log'] = details_str
        context['percentage'] = percentage

```

```
return render(request, 'q1_t.html', context)
```

Output:

127.0.0.1:8000/lab5/q1/

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Student Records System

Enter Student Details

Student Name

Date of Birth

Address

Contact Number

Email ID

Academic Marks

English

Physics

Chemistry

Submit Details

Submission Output

Formatted Details:

Name: Adi
Address: Block 14, MIT Manipal
Email: adi022832@gmail.com
Marks: Eng(88), Phy(98), Chem(60)

Total Percentage: **82.0%**

Q2. Create a web form with employee ids in the dropdown list. Use a textbox for accepting the date of joining. Add a button named “Am I Eligible for Promotion”. On clicking the button, if he has more than 5 years of experience, then display “YES” else “NO” in the label control.

Template:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Promotion Eligibility Checker</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap
.min.css" rel="stylesheet">
  <style>
    body { background-color: #f8f9fa; }
    .result-box {
      font-size: 1.5rem;
      font-weight: bold;
      min-height: 60px;
    }
  </style>
</head>
<body>

<nav class="navbar navbar-expand-lg navbar-dark bg-dark mb-4">
  <div class="container">
    <a class="navbar-brand" href="#">Lab 5</a>
    <div class="navbar-nav">
      <a class="nav-link" href="/lab5/q1/">Q1: Student Form</a>
      <a class="nav-link" href="/lab5/q2/">Q2: Promotion Check</a>
    </div>
  </div>
</nav>

<div class="container py-5">
  <div class="row justify-content-center">
    <div class="col-md-6">
      <div class="card shadow">
        <div class="card-header bg-success text-white">
          <h5 class="mb-0">HR Portal: Promotion Check</h5>
        </div>
        <div class="card-body">
```

```

<form method="POST">
    {% csrf_token %}

    <div class="mb-3">
        <label for="emp_id"
class="form-label">Select Employee ID:</label>
        <select class="form-select" name="emp_id"
required>
            <option selected disabled
value="">Choose an ID...</option>
            <option
value="EMP001">EMP001</option>
            <option
value="EMP002">EMP002</option>
            <option
value="EMP003">EMP003</option>
            <option
value="EMP004">EMP004</option>
        </select>
    </div>

    <div class="mb-3">
        <label for="doj" class="form-label">Date
of Joining:</label>
        <input type="date" class="form-control"
name="doj" required>
    </div>

    <div class="d-grid">
        <button type="submit" class="btn
btn-success">Am I Eligible for Promotion?</button>
    </div>
</form>

{% if result %}
<hr class="my-4">
<div class="text-center">
    <p class="text-muted mb-1">Eligibility
Status:</p>
    <span class="badge rounded-pill
{% if result == 'YES' %}bg-success{% else
%}bg-danger{% endif %}
result-box d-inline-flex
align-items-center px-4">
        {{ result }}

```



```

        </span>
    </div>
    {% endif %}

</div>
</div>
</div>
</div>
</div>

</body>
</html>

```

Backend:

```

def employee_view(request):
    result = None

    if request.method == 'POST':
        doj_str = request.POST.get('doj')

        if doj_str:
            doj = datetime.strptime(doj_str, "%Y-%m-%d").date()
            today = date.today()

            delta = today - doj
            years_experience = delta.days / 365.25

            if years_experience > 5:
                result = "YES"
            else:
                result = "NO"

        return render(request, 'q2_t.html', {'result': result})

```

Output:

127.0.0.1:8000/lab5/q2/

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Lab 5

Q1: Student Form Q2: Pr

HR Portal: Promotion Check

Select Employee ID:

Choose an ID...v

Date of Joining:

dd/mm/yyyy

Am I Eligible for Promotion?

Eligibility Status:

NO